Ship Knowledge and Equipment

NTA LEVEL 4 & BRIDGE COURSE

SRT 04104

Basic fire prevention and fire fighting (BFPFF)




HYUNDAI

Lecture contents

- Introduction
- Fire/Combustion
- FPFF
- Types of combustion
- Modes for spread of fire
- Rate of combustion
- Important definitions in fire concept

Introduction

Fire and explosions are the greatest potential hazard to an off shore/on shore installation. So all personnel involved in these operations need to have an understanding of the concepts of fire, fire ignition temperatures, fire prevention and fire fighting.

- Fire

It is a chemical chain reaction which takes place with the evolution of heat and light.

Fire is the rapid oxidation of a material in the exothermic chemical process of combustion, releasing heat, light, and various reaction products.

Various Reaction products

- Gases: result from the modification in the composition of the combustible.
- Smoke: appears due to incomplete combustion, in which small particles become visible, varying in colour, size and quantity, and sometimes soot.
- Flames: the most visible result of combustion. It is a zone of incandescent gases visible around the surface of the material in combustion. Flames are no more than the combustion of gas.

Fire Prevention

- Fire safety/prevention refers to precautions that are taken to prevent or reduce the likelihood of a fire that may result in death, injury, or property damage.
- Firefighting is the act of extinguishing fires. A firefighter suppresses and extinguishes fires to prevent loss of life, and/or destruction of property and the environment. Firefighting is a highly technical skill that requires professionals who have spent years training in both general firefighting techniques and specialized areas of expertise.

Types of combustion

I. Slow or incipient combustion- in which the amount of heat & light emitted is feeble.

II. Rapid or active combustion- in which a considerable amount of heat & light is emitted within a short time.



Types of combustion…

iii. Deflagration - which takes place with a considerable rapidity, evolving heat & light.



Types of combustion…

iv. Explosion - is a very rapid combustion with a loud noise within an extremely short time with generation of very high pressure & temperature.



Modes for spread of fire

- Conduction

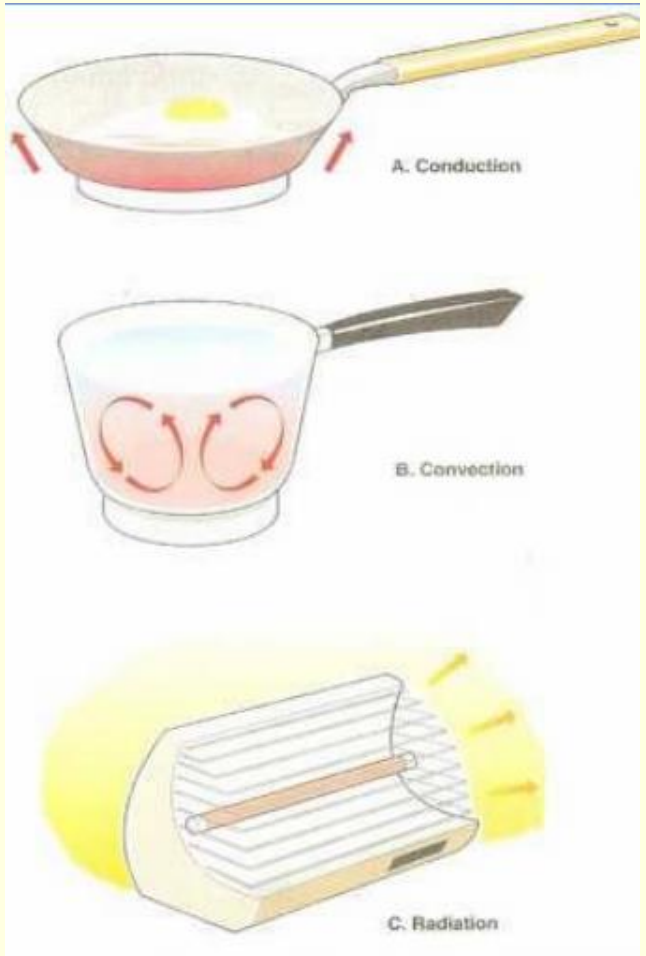
It occurs only in solids i.e. metallic objects.

- Convection

It occurs in both in liquids & gases.

- Radiation

It is neither conduction nor convection. These are heated rays emanating from the hot object.



Rate of Combustion

A rate of combustion or the spread of fire would depends on :

- The area of solid/liquid in contact with air.
- The amount of heat generated to raise the Temperature of un-burnt portion.
- The ability of materials to conduct heat away.
- Atmospheric humidity.
- Wind velocity.
- Temperature.

Important Definitions

- **Flash Point** : It is the lowest temperature at which an inflammable substance gives off sufficient vapours, so as to form a momentary flash on application of a pilot flame.
- **Fire Point** : It is the lowest temperature at which the heat from the combustion of burning vapours is capable of producing sufficient vapours to enable combustion to continue.

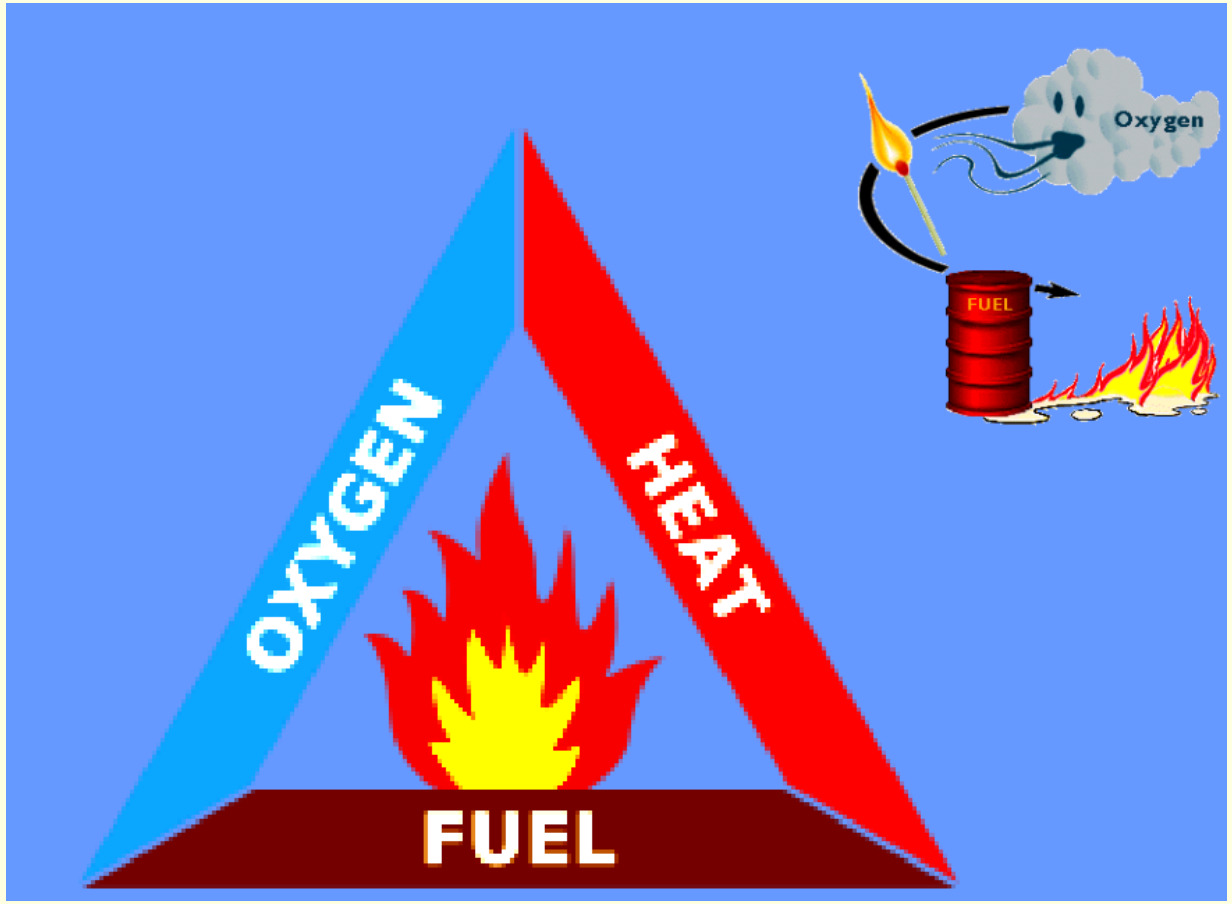
The Fire Point is generally above the Flash Point.

Important Definitions…

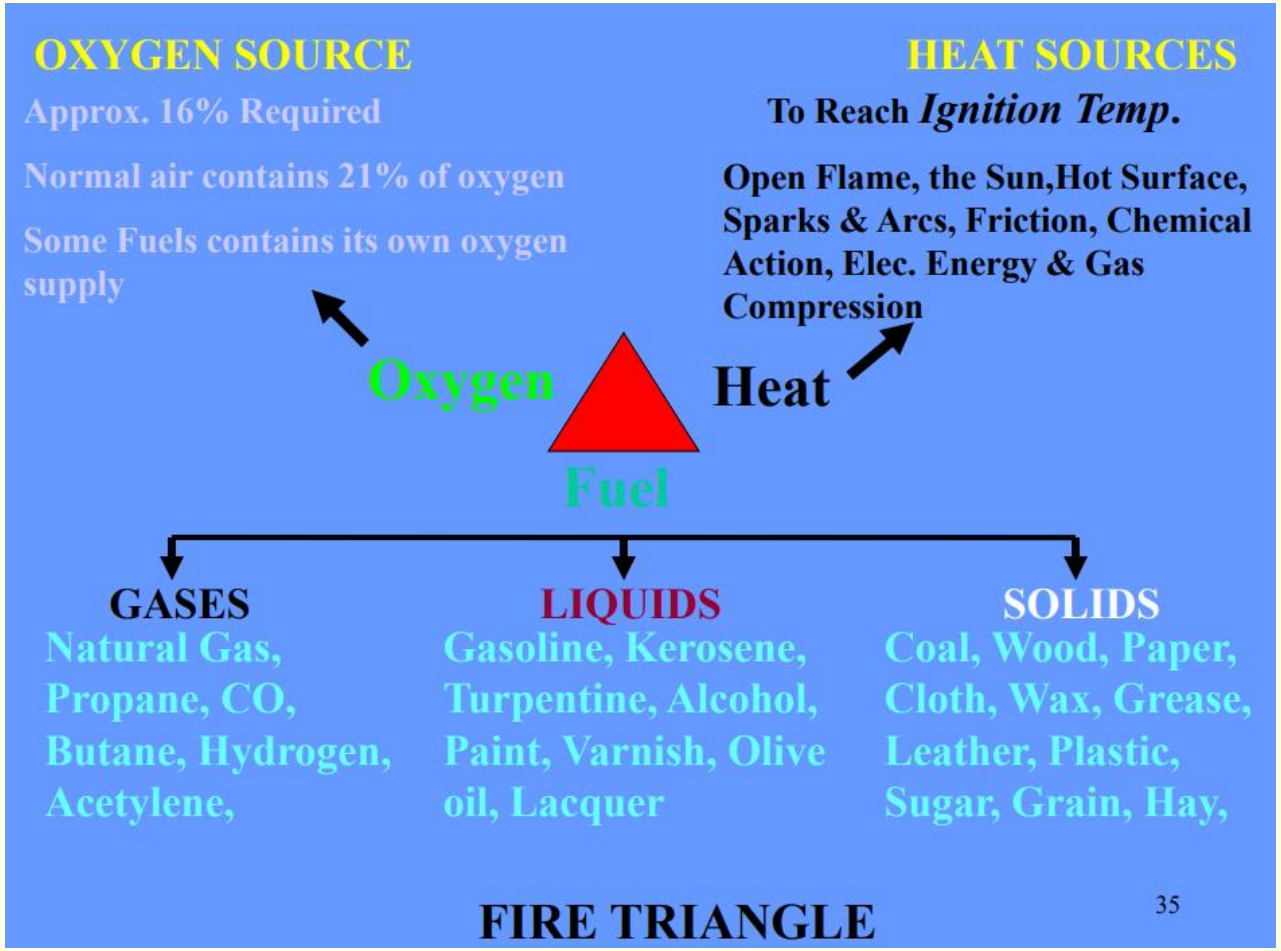
- **Ignition Temperature** : It is the lowest temperature at which spontaneous combustion can takes place without application of an external heat.
- **Auto Ignition Point** : Refers to the temperature to which a substance must reach, before it ignite, in the absence of flame, but in presence of air.
- **Spontaneous Combustion** : It occurs as a result of heat generated by the reacting substances without any external heat.

  e.g. Hot glycerine + Potassium Permagnate

Fire Triangle



Fire Triangle…



Classes of fire

- Fire is categorized into 5 classes namely class A,B,C, D and E.
- **Class A** fire is a Solid Combustibles (generally of organic origin examples Wood, Coal, Paper, Cloth and Straw.
- **Class B** fire is a Liquid Combustibles or easily Liquefied Solids such as Gasoline, Varnishes, Wax, Alcohol and Tar.
- **Class C** fire is Gaseous Combustibles such as Butane, Propane, Acetylene, Hydrogen and Hexane.
- **Class D** fire is a Metal Combustibles such as Magnesium, Aluminium, Sodium, Uranium and Lithium.

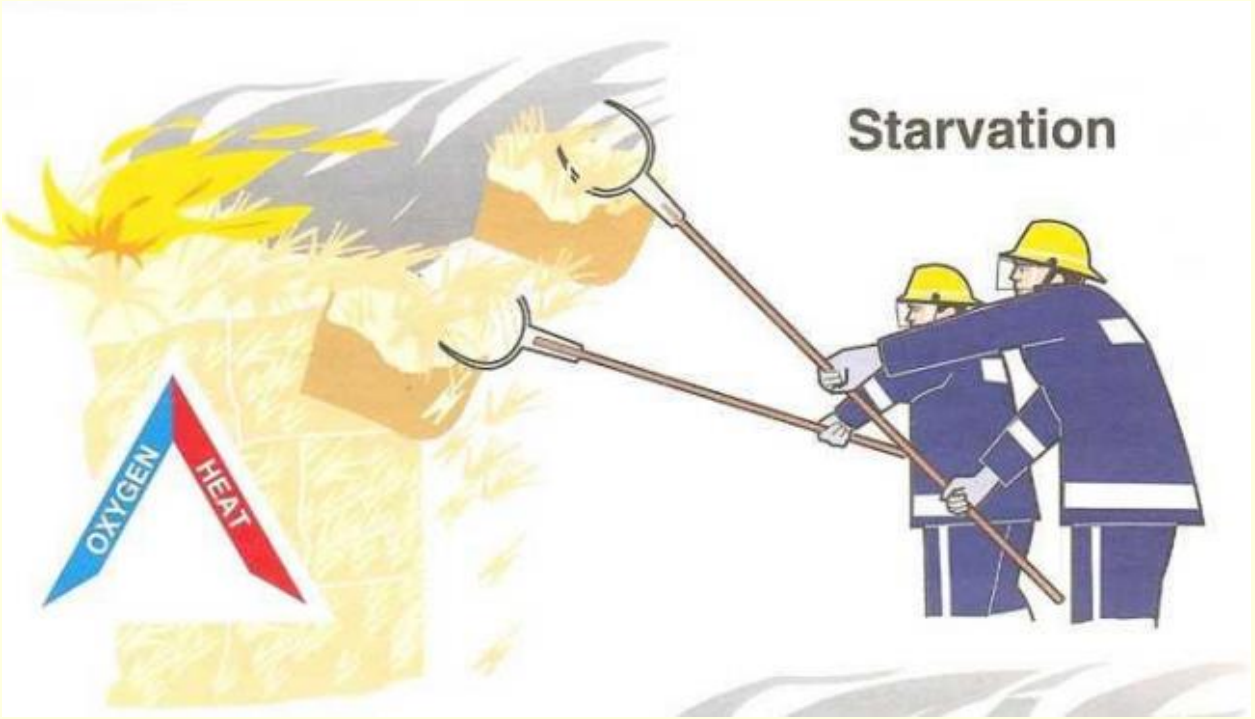
Classes of fire…

- **Class E** fire is an Electrical fires. Fires complicated by the presence of electricity.
- The five mentioned classes are going to facilitate the selection of the most adequate strategy applicable to extinguish a certain fire.
- Each category requires an appropriate way of extinction and method.

Principles of Fire Extinction

- **Starvation**

Removal of fuel material from fire area.



Principles of Fire Extinction…

- **Smothering**

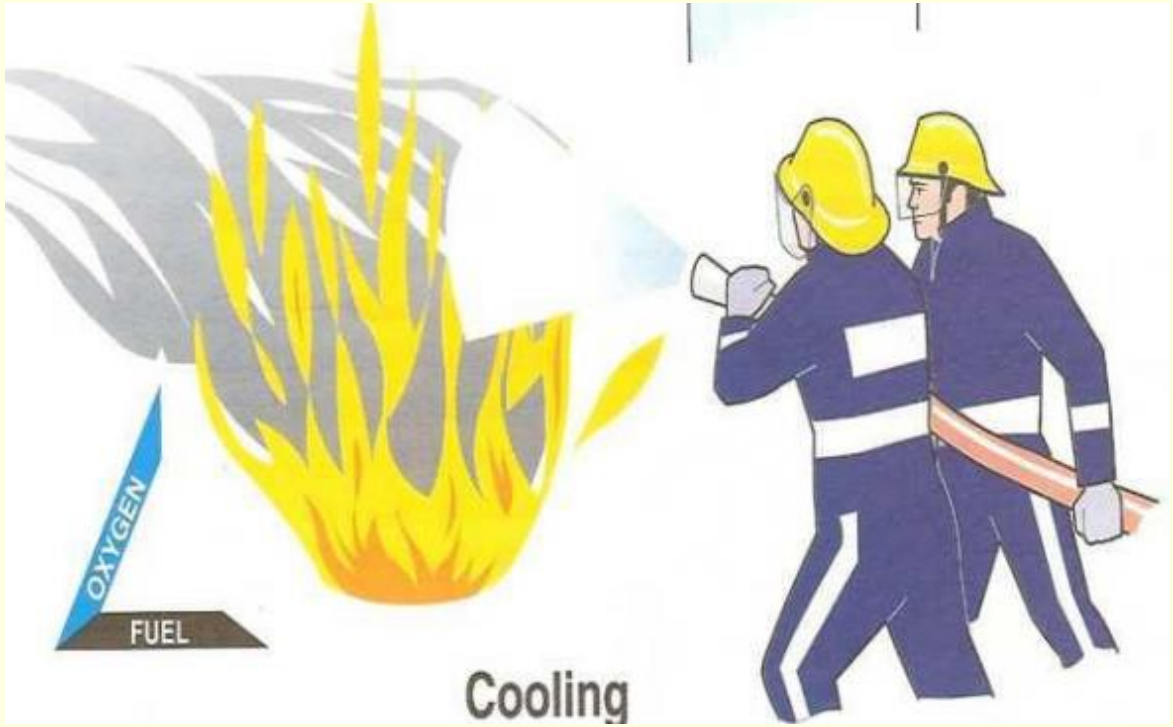
Cutting off the supply of oxygen from fire area.



Principles of Fire Extinction…

- **Cooling**

Removal of heat from the burning material/fire area.



Extinguishing agents

- Generally there are five types of extinguishing agents used in fire fighting as follows,
  i. CO_2
  ii. Foam
  iii. Water
  iv. Dry chemical Powder
  v. Blanket and
  vi. Halons (prohibited)

Classes of fire and their respective extinguishing agents.

- Class A fire we use water, chemical powder and CO_2.
- Class B fire we use Foam, Dry chemical powder and CO_2.
- Class C fire we use CO_2, and Dry chemical Powder.
- Class D fire we use Dry chemical powder only.
- Class E fire we use Dry chemical powder, CO_2, halon.

- Make sure that you use a correct extinguisher to a correct class of fire, inversely of this will results to,
  - Fire will not be extinguished.
  - Explosion.
  - Fire will be accelerated, and Spreading of fire.

Colours for extinguishing agents

- Red– water



- Black– CO_2



- Blue – Dry chemical powder



- Cream – Foam
- Green– halon [prohibited].

Fire fighting appliance

- Fire hoses and nozzles
- Mobile and non movable apparatus
  - ✓ CO_2
  - ✓ Foam
  - ✓ Sprinklers
  - ✓ Dry chemical powder
  - ✓ Fire mains and hydrants

Fire fighting appliance…

- O_2 for breathing.
- Fireman's outfit.
- Axe.
- Helmet.
- Overall,
- Gloves,
- Safety boot,
- Safety rope,

Fire fighters





Areas of fire hazards on board the ship

- **Fire in the Machinery space**:-causes, including
  - ✓ Combustible liquids leaking through faulty or damaged connections.
  - ✓ Hot surface e.g exhaust pipes, engine parts overheating in close proximity to oil lines.
  - ✓ Hot work e.g welding, cutting.
  - ✓ Auto ignition, e.g oil dripping on hot surface.

Areas of fire hazards on board the ship…

- **Fire in accommodation** causes , including:
  - ✓ Combustible materials
  - ✓ Cigarette smoking, including careless disposal of burning cigarettes or ash
  - ✓ Textiles adjacent to hot objects such as radiators and lamps
  - ✓ Defective and overloaded electrical systems.
  - ✓ In a laundry, incorrect installation of tumble drier or failure to keep it clean
  - ✓ Electrical equipment about 90% of marine fire caused by faulty of electrical equipment due to lack of maintenance and incorrect use of electrical appliance.

Areas of fire hazards on board the ship…

- **Fire in holds and in containers**; causes including
  - ✓ Cargoes liable to self-heating and spontaneous combustion (coal), bulk cargoes liable to emit flammable gas (coal).
  - ✓ Loss of integrity of packages containing explosive, flammable or reactive substances.
  - ✓ Collection of oily materials as a result of insufficient cleaning and of leakage form tanks.
- **Fire in the galley causes**, including
  - ✓ Overheating of combustible liquids and fats, overheating of deep-fat fryers, hot surface.

Any Questions, Suggestions, Comments, etc.

THE END

